

2026 春. 复变

一.

1. 绝对收敛

$$2. \frac{2z}{(1-z)^3}$$

$$3. 0 < |z| < \frac{1}{5}$$

$$4. -i\pi e^{\pi^2}$$

$$5. \pm \ln 6$$

$$6. -3 \times 2024!$$

$$\therefore f(z) = \frac{1}{(3z^3)^2} \cdot \frac{1}{\left(1 + \frac{1}{3z^3}\right)^2}, \quad |t| = \left|\frac{1}{3z^3}\right| \in \left(\frac{1}{6\sqrt{2}}, \frac{2\sqrt{2}}{3}\right) < 1.$$

$$\text{令 } t = \frac{1}{3z^3}.$$

$$\begin{aligned} \frac{1}{(1+t)^2} &= -\left(\frac{1}{1+t}\right)' = -\left[\sum_{n=0}^{\infty} (-1)^n t^n\right]' \\ &= \sum_{n=0}^{\infty} (-1)^n (n+1) t^n \end{aligned}$$

$$\begin{aligned} f(z) &= t^2 \sum_{n=0}^{\infty} (-1)^n (n+1) t^n = \sum_{n=0}^{\infty} (-1)^n (n+1) t^{n+2} \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n (n+1)}{3^{n+2}} \cdot \frac{1}{z^{3n+6}} \end{aligned}$$

三.

1. $f(z)$ 在 $\mathbb{C}$ 上奇点有 $0, 1, k\pi (k \neq 0, k \in \mathbb{Z}), \infty$

① $z=0$ 可去奇点

$$\lim_{z \rightarrow 0} f(z) = \lim_{z \rightarrow 0} \left(\frac{1}{z} - \frac{1}{\sin z} \right) = 0 \in \mathbb{C}$$

② $z=1$ 本性奇点.

$z \cos \frac{z}{z-1}$ 展开后无穷多负幂项

$\lim_{z \rightarrow 1} f(z)$ 不存在.

③ $z=k\pi (k \neq 0, k \in \mathbb{Z})$ 简单极点.

为 $\sin z$ 1级零点.

④ $z=\infty$ 非孤立奇点

奇点 $k\pi$ 逼近 ∞ .

2. 仅 $z \cos \frac{z}{z-1}$ 对负幂项贡献.

替换 $x = z-1$.

$$\Rightarrow (x+1) \cos \left(1 + \frac{1}{x} \right) = (x+1) \left[\cos 1 \cos \frac{1}{x} - \sin 1 \sin \frac{1}{x} \right]$$

前项贡献 $x \cdot \cos(1) \cdot \frac{-1}{2026!} \cdot \frac{1}{x^{2026}}$

后项贡献 $1 \cdot (-\sin(1)) \cdot \frac{1}{2025!} \cdot \frac{1}{x^{2025}}$

$$\text{则 } C_{-2025} = -\frac{\cos(1)}{2026!} - \frac{\sin(1)}{2025!}$$

3.

$$\textcircled{1} z=1.$$

如 2 中展开. 得. 负一项为

$$x \cdot \cos 1 \left(-\frac{1}{2!}\right) \cdot \frac{1}{x^2} + 1 \cdot (-\sin 1) \cdot \frac{1}{x}$$

$$= \left[-\frac{1}{2} \cos(1) - \sin(1)\right] \frac{1}{x}.$$

$$\underline{\text{Res}[f(z), 1] = -\frac{1}{2} \cos(1) - \sin(1)}.$$

$$\textcircled{2} z = k\pi. (k \neq 0, k \in \mathbb{Z})$$

$$\text{Res}[f(z), k\pi] = \text{Res}\left[-\frac{1}{\sin z}, k\pi\right]$$

$$= -\frac{1}{\cos(k\pi)} = \underline{(-1)^{k+1}}$$

四.

$$I = \frac{1}{2} \int_{-\infty}^{+\infty} \frac{\cos(2x) + 1}{x^2 - 4x + 5} dx$$

$$I_0 = \int_{-\infty}^{+\infty} \frac{1}{x^2 - 4x + 5} dx \quad \text{被积函数奇点 } 2 \pm i$$

$$= 2\pi i \text{Res}\left[\frac{1}{x^2 - 4x + 5}, 2+i\right]$$

$$= 2\pi i \cdot \frac{1}{2i} = \pi$$

$$I_1 = \int_{-\infty}^{+\infty} \frac{\cos(2x)}{x^2 - 4x + 5} dx$$

$$I_2 = \int_{-\infty}^{+\infty} \frac{\sin(2x)}{x^2 - 4x + 5} dx,$$

$$J = I_1 + iI_2 = \int_{-\infty}^{+\infty} \frac{e^{i \cdot 2x}}{x^2 - 4x + 5} dx$$

奇点 $2 \pm i$
 e 指数系数
 $2 > 0$

$$= 2\pi i \operatorname{Res} \left[\frac{e^{i \cdot 2x}}{x^2 - 4x + 5}, 2 + i \right]$$

$$= 2\pi i \cdot \frac{e^{-2+4i}}{2i}$$

$$= \frac{\pi}{e^2} \cdot e^{4i}$$

$$I_1 = \operatorname{Re}(J) = \frac{\pi}{e^2} \cos(4)$$

$$I = \frac{1}{2}(I_0 + I_1)$$

$$= \frac{\pi}{2} + \frac{\pi}{2e^2} \cos(4)$$

五. 被积函数 $\bar{C}$ 上奇点 $0, 1, 3, \infty$. 均孤立.

$0, 1$ 在 $|z| < 2$ 内. $f(z) = \frac{z^2 \sin \frac{1}{z}}{(z-1)(z-3)}$

由留数定理.

$$\text{原式} = -2\pi i \left[\text{Res}[f(z), 3] + \text{Res}[f(z), \infty] \right]$$

3 为简单极点

$$\text{Res}[f(z), 3] = \frac{9}{2} \sin \frac{1}{3}$$

∞ 处. ~~Res~~

$$- \frac{f(\frac{1}{z})}{z^2} = - \frac{\sin z}{(1-z)(1-3z)z^2}, \quad 0 \text{ 为简单极点.}$$

$$\begin{aligned} \text{Res}[f(z), \infty] &= \text{Res}\left[-\frac{f(\frac{1}{z})}{z^2}, 0\right] \\ &= -1. \end{aligned}$$

$$\text{原式} = -2\pi i \left[\frac{9}{2} \sin \frac{1}{3} - 1 \right]$$

$$= \underline{2\pi i - 9\pi i \cdot \sin \frac{1}{3}}$$

5.

$$1. K = \{ f(z) = 2+i, f(z) = 2-i, \\ f(z) = -2+i, f(z) = -2-i \}$$

其中 $f(z)$ 均定义在 T 上, 0 处无定义.
常函数.

2. 结合 Liouville 定理.